Power input, USB–C and 3.3 V

File: 01_Power_Input_USB.kicad_sch

ESP32–S3, native USB, reset and boot

File: 02_ESP32_USB_Reset.kicad_sch

E–paper power and interface

File: 03_Display_EPaper.kicad_sch

Touch always–on rail and RTC

File: 04_Touch_RTC.kicad_sch

AUTOTERM UART and 1–Wire

File: 05_AUTOTERM_1Wire.kicad_sch

External controls, lighting and diagnostics

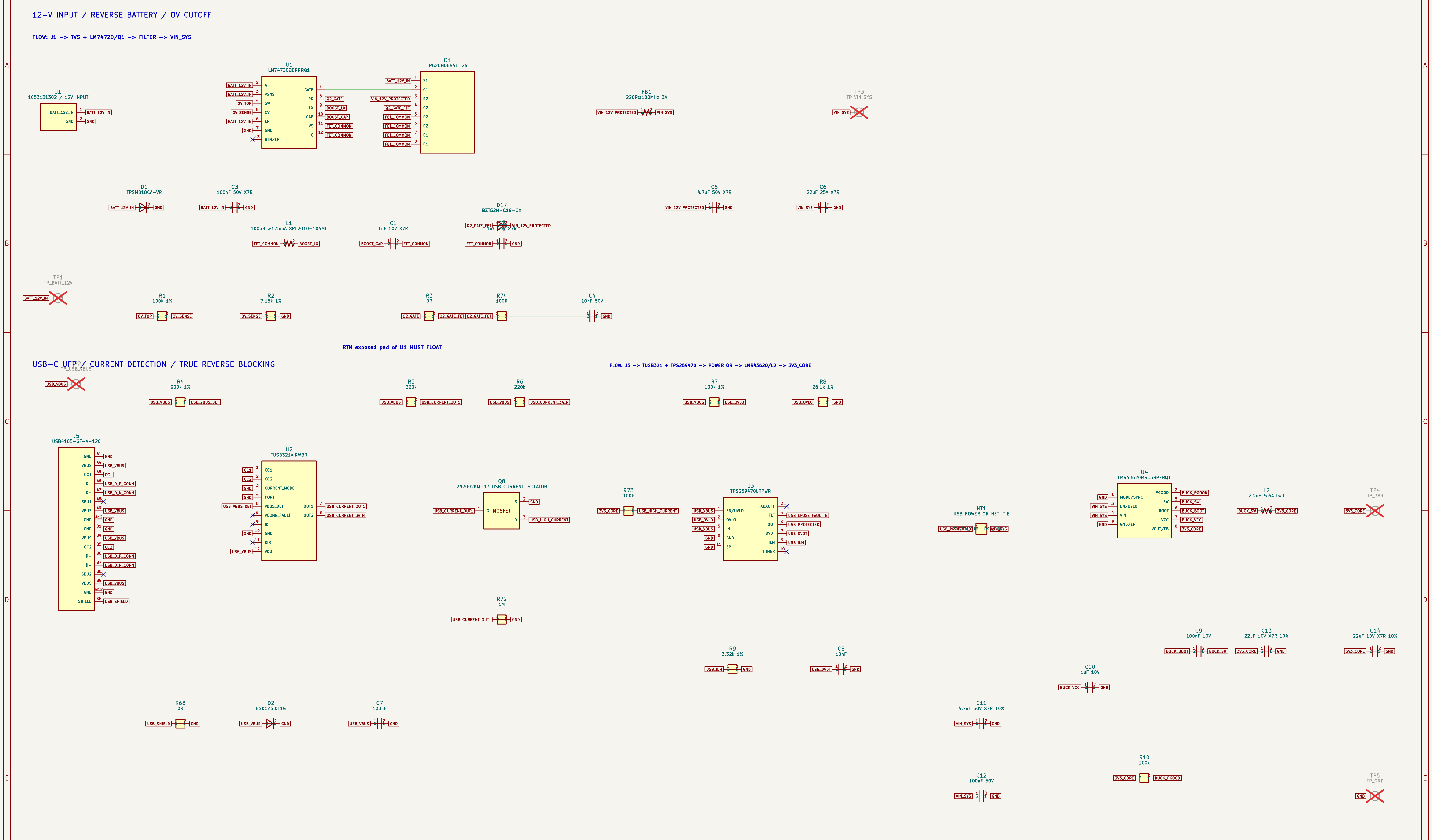
File: 06_Controls_Diagnostics.kicad_sch

ESP32–S3–WROOM–1U–N16R8 / 10–14V vehicle auxiliary battery / USB–C
Review schematic – NOT RELEASED FOR PRODUCTION
LandyHeater

Sheet: /
File: LandyHeater–Board.kicad_sch

Title: Landy Heater Controller – Hierarchy

Size: A4	Date: 2026–09–06	**Rev: A / Phase 2**
KiCad E.D.A. 10.0.6		Id: 1/7



12 V protected input | USB-C sink and reverse blocking | 3.3 V automotive buck
Review schematic - NOT RELEASED FOR PRODUCTION
LandyHeater
Sheet: /Power input, USB-C and 3.3 V/
File: 01_Power_Input_USB.kicad_sch

Title: Power input, USB-C and 3.3 V

Size: A3 | Date: 2026-09-06 | Rev: A / Phase 2
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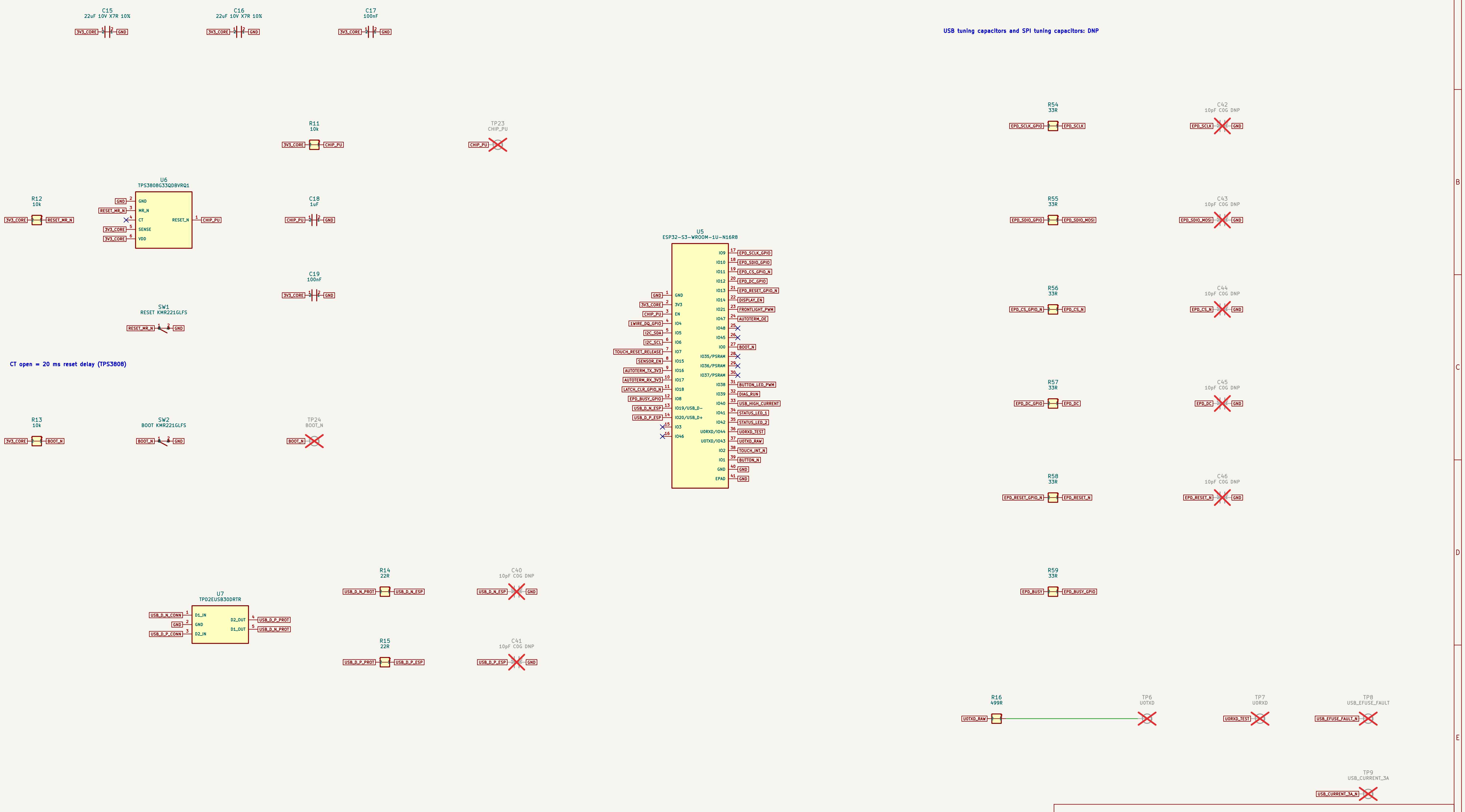
ESP32-S3-WROOM-1U-N16R8 / SAFE RESET AND BOOT

FLOW: 3V3 -> TPS3808/RESET -> CHIP_PU | USB J5 -> ESD -> R14/R15 -> GPIO19/20 | GPIO -> SPI SERIES PARTS



N16R8 octal-PSRAM pins reserved | Native USB GPIO19/20 | Manual BOOT and RESET

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Sheet: /ESP32-S3, native USB, reset and boot/

File: 02_ESP32_USB_Reset.kicad_sch

Title: ESP32-S3, native USB, reset and boot

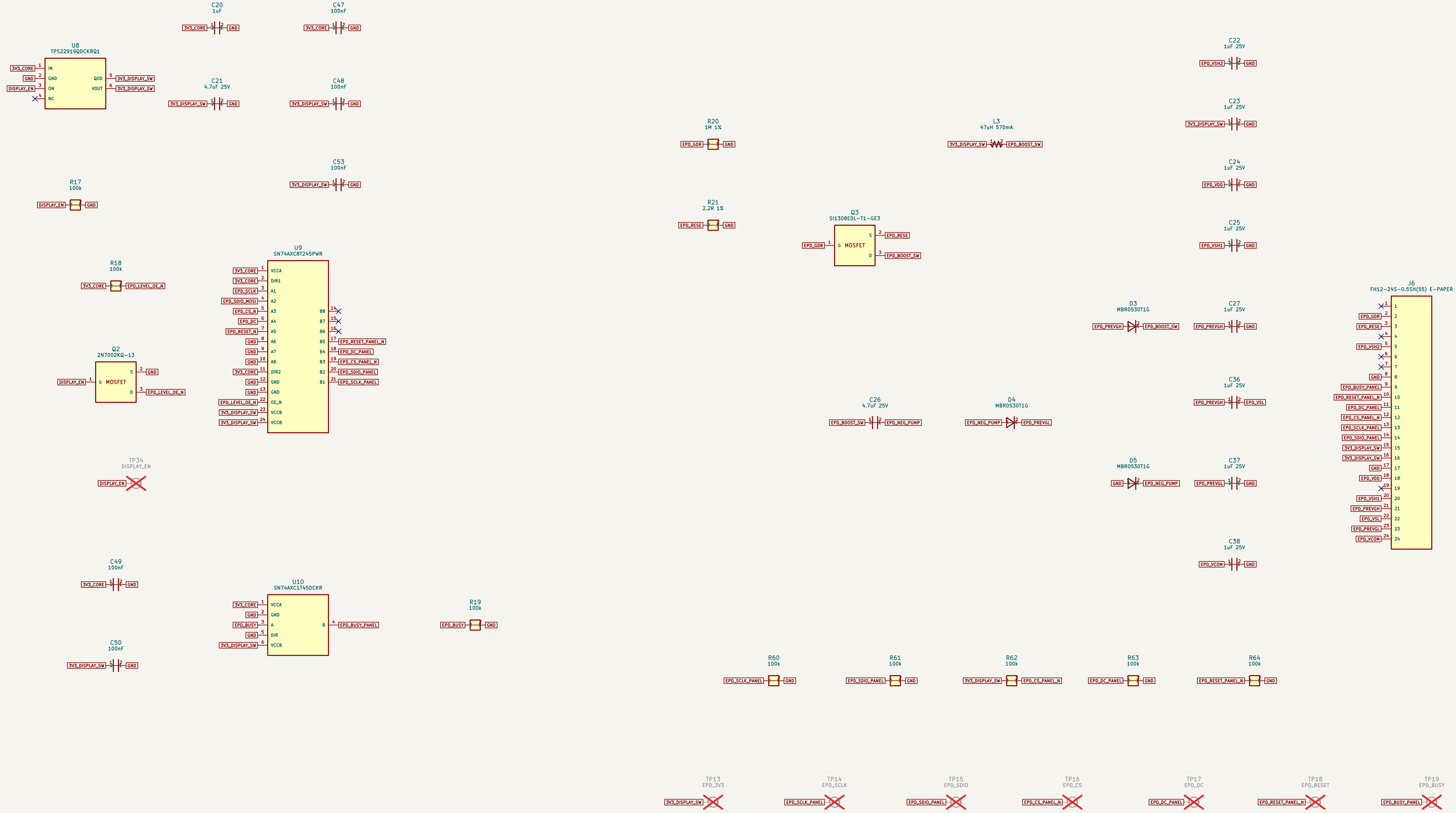
Size: A3 | Date: 2026-09-06 | Rev: A / Phase 2

KiCad E.D.A. 10.0.6 | Id: 3/7

DISPLAY POWER-OFF ISOLATION

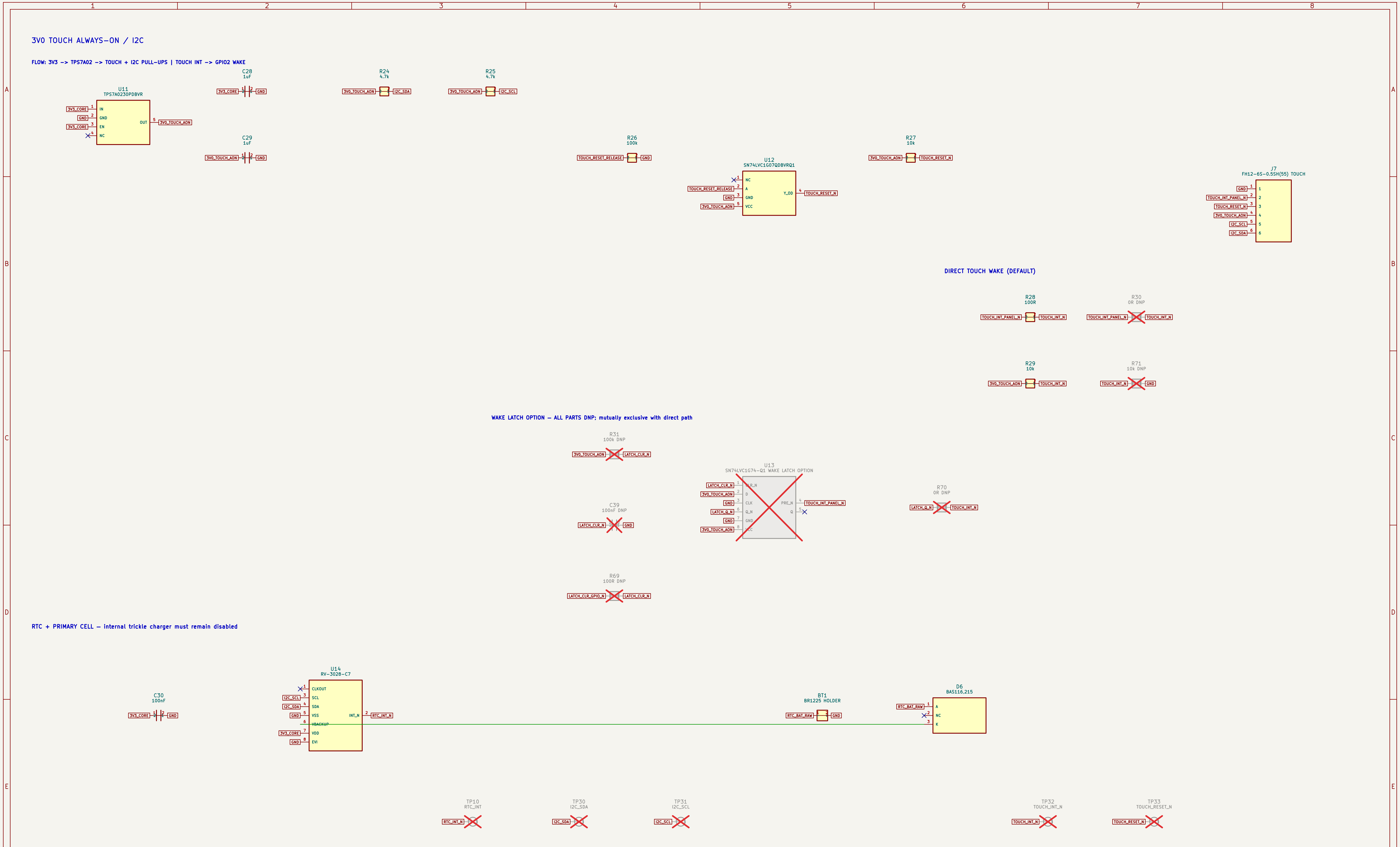
GOOD DISPLAY CHAPTER 12 BOOSTER — DO NOT SUBSTITUTE TOPOLOGY

FLOW: GPIO SPI -> AXC LEVEL/POWER-OFF ISOLATION -> J6 | 3V3_DISPLAY_SW -> BOOSTER -> PANEL RAILS



FPC contact side / pin-1 orientation: NOT VERIFIED until original sample inspection

Write-only SPI | Power-off isolation | FPC orientation remains a physical sample gate
Review schematic — NOT RELEASED FOR PRODUCTION
LandyHeater
Sheet: /E-paper power and interface/
File: 03_Display_EPaper.kicad_sch
Title: E-paper power and interface
Size: A3 | Date: 2026-09-06 | **Rev: A / Phase 2**
KiCad E.D.A. 10.0.6 | Id: 4/7



Touch can wake in comfort standby | Wake-latch option DNP pending pulse measurement | BR1225 i

Review schematic – NOT RELEASED FOR PRODUCTION

LandyHeater

Sheet: /Touch always-on rail and RTC/

File: 04_Touch_RTC.kicad_sch

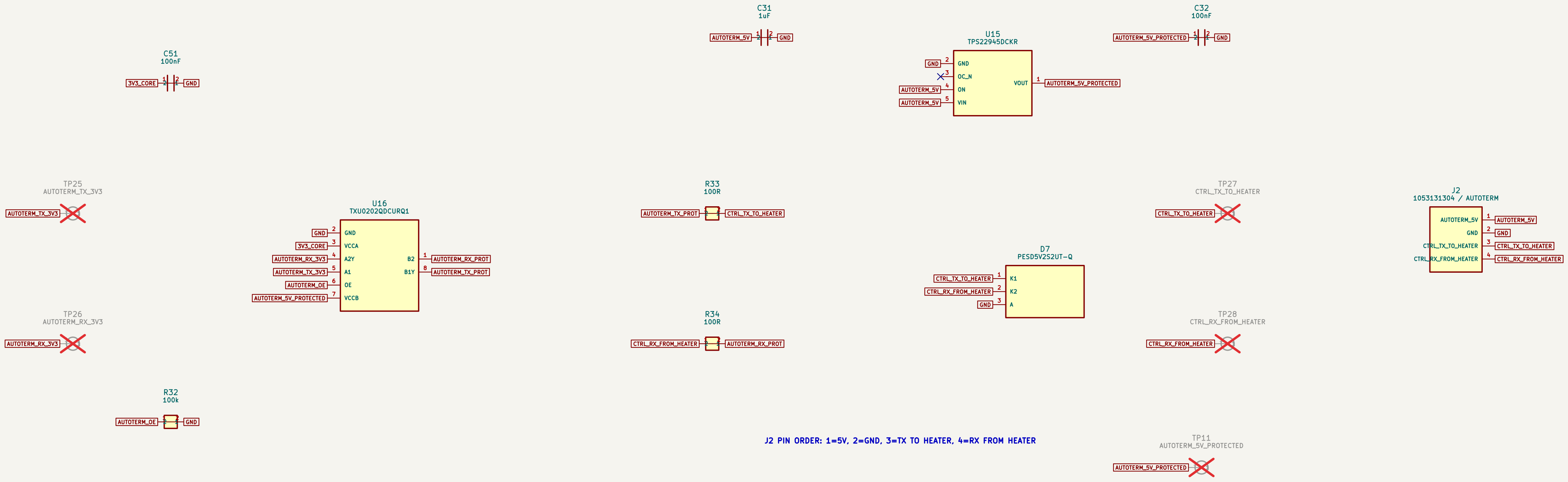
Title: Touch always-on rail and RTC

Size: A3 | Date: 2026-09-06 | **Rev: A / Phase 2**

KiCad E.D.A. 10.0.6 | Id: 5/7

AUTOTERM UART — 5-V side powered only by heater

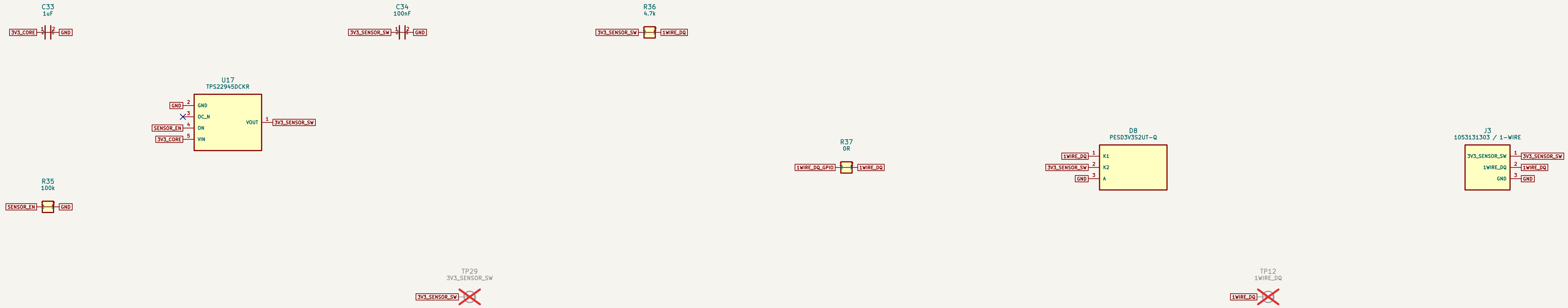
FLOW: ESP32 UART -> TXU0202 -> SERIES + ESD -> J2



J2 PIN ORDER: 1=5V, 2=GND, 3=TX TO HEATER, 4=RX FROM HEATER

1—WIRE — THREE DS18B20 / 2-5 m combined harness

FLOW: 3V3 -> TPS22945 -> SENSOR SUPPLY | GPIO4 -> 0R + ESD -> J3



Heater-side 5 V powers level shifter B side | Three parallel DS18B20 on one 2-5 m trunk
Review schematic - NOT RELEASED FOR PRODUCTION

LandyHeater

Sheet: /AUTOTERM UART and 1-Wire/
File: 05_AUTOTERM_1Wire.kicad_sch

Title: AUTOTERM UART and 1-Wire

Size: A3	Date: 2026-09-06	**Rev: A / Phase 2**
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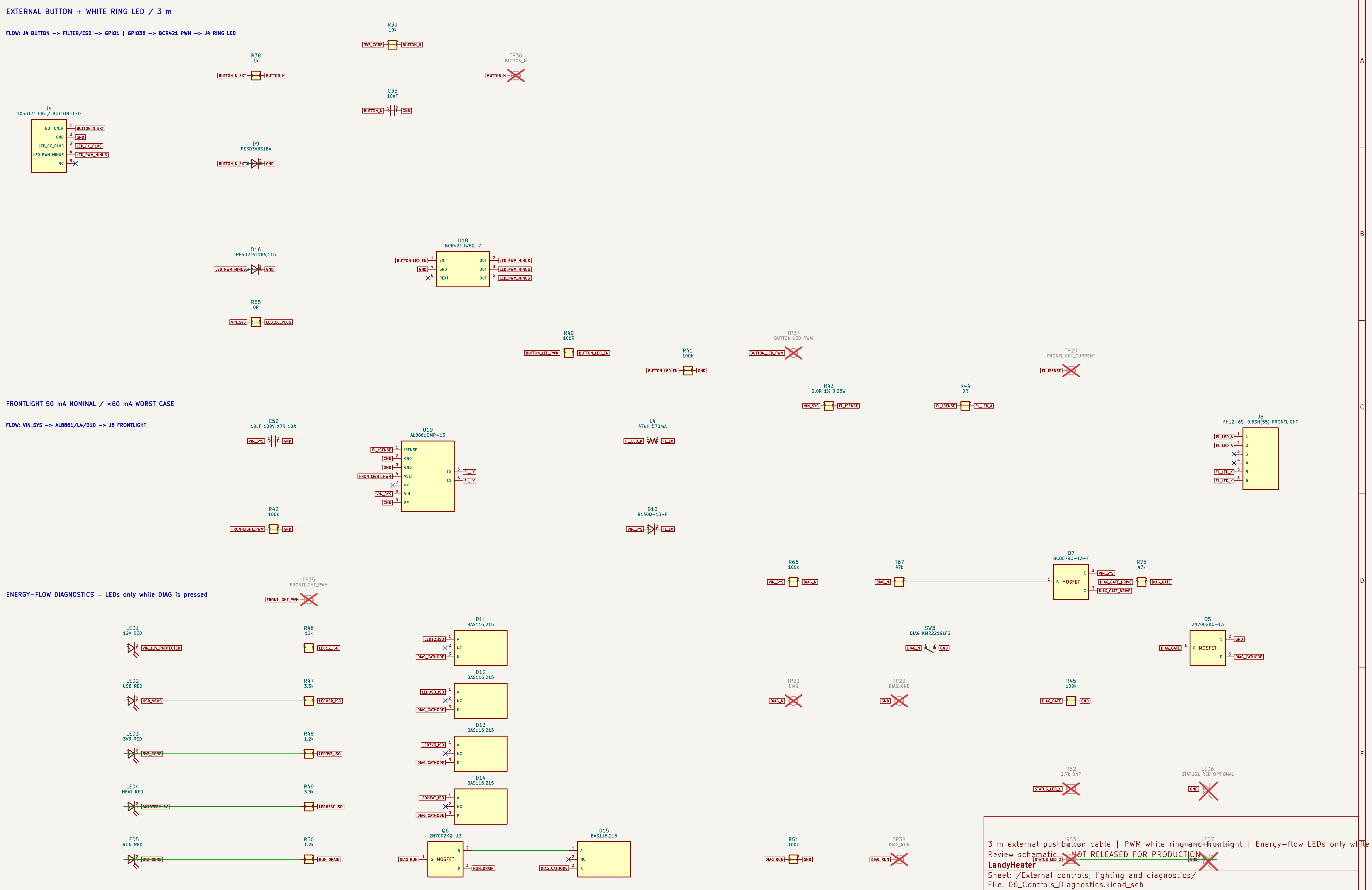

3 m external pushbutton cable | PWM white ring and frontlight | Energy-flow LEDs only while DIAG is
Review schematic – NOT RELEASED FOR PRODUCTION
LandyHeater

Sheet: /External controls, lighting and diagnostics/
File: 06_Controls_Diagnostics.kicad_sch

Title: External controls, lighting and diagnostics

Size: A3	Date: 2026-09-06	**Rev: A / Phase 2**
KiCad E.D.A. 10.0.6		Id: 7/7